Matter In Our Surroundings

Matter: Anything that has mass and occupies space is called matter.
The force of attraction between particles can be arranged in decreasing order as: Solid > Liquid > Gas.

Solid

Solid is defined as that form of matter which possesses rigidity, is incompressible and hence has a definite shape and a definite volume.

Liquid

Liquid is defined as that form of matter which possesses fluidity but is almost incompressible and hence has a definite volume but no definite shape.

Gas

Gas is defined as that form of matter which possesses fluidity but it is highly compressible and hence has neither definite shape nor definite volume.

Characteristics of particles:

- Large number of particles constitute matter.
- Particles of matter are very small in size.
- Particles of matter have spaces between them.
- Particles of matter are continuously in motion
- Particles of matter attract each other.

Density:

It is the mass occupied by a solid per unit volume and is obtained by dividing the mass of a particular solid by the volume occupied.

Melting point:

The melting point of a solid may be defined as the temperature at which a solid melts to become a liquid at atmospheric pressure.

Boiling Point:

The temperature at which a liquid starts boiling at the atmospheric pressure is known as the boiling point.

Freezing point:

The temperature at which a liquid changes to solid by giving out heat at the atmospheric pressure.

Transpiration:

The process of evaporation of water from the aerial parts of plants especially leaves is called transpiration.

S.I. unit of temperature is Kelvin.:

- T (K) = t (°C) + 273.
- The melting point of ice is 273.16 K (0°C)
- The boiling point of water is 373 K (100° C).

Sublimation:

It is the change of a solid directly into the gaseous state without passing through the liquid state upon heating and back to the solid state when the temperature is lowered.

Latent heat of vaporisation:

It is the heat energy required to change 1 kg of liquid to gas at atmospheric pressure at its boiling point. Boiling is a bulk phenomenon.

Humidity:

The amount of water vapour present in the air.

Evaporation:

The change of liquid to a vapour state at any temperature below the boiling point of the liquid is termed as evaporation.

Latent heat of fusion:

The amount of heat energy that is needed to convert 1 kg of a solid into the liquid state at atmospheric pressure at its melting point is termed as latent heat of fusion.